ANSWER KEYS - PROFESSOR COPY

Week 1 Lecture Quizzes and Final

Freshman Seminar — Week 1 (pass mark 6/8)

1. D - Lectures teach; recitations practice. At NICUniversity, recitation is where you do the homework for the lecture you just had.

2. C - Three big ideas, in your own words. Notes are for future-you, and future-you needs the ideas, not a copy of the whole lecture.

3. C - You can copy words without understanding them. You can't rewrite an idea in your own words without understanding it — so the rewriting IS the learning.

4. C - Every medical school she applied to turned her down until Geneva Medical College in New York said yes — partly as a joke. She graduated first in her class.

5. D - Read them all, cross out the wrong ones, then choose. (And at NICUniversity, the right answer is placed at random — so length and position are no help. You have to know it!)

6. B - Pre-medical: the science and math courses (chemistry, biology, physics, statistics…) you take as an undergraduate so you're ready for medical school afterward.

7. B - Every lecture assigns homework at the end, and each homework gets its own recitation slot — usually the next morning. (Freshman Seminar's is the exception: its recitation is Monday afternoon.)

8. A - Your transcript is your official record — courses, grades, credits. (Tricky one: 'transcript' can also mean a written copy of speech, but at a university it means your record.)

Chemistry 101 — Week 1 (pass mark 6/8)

1. A - Atoms are the building blocks of all ordinary matter — solids, liquids, and gases; living and non-living. (Tiny living creatures are cells or bacteria — that's Biology's job.)

2. D - A subscript counts the atom right before it. H_2O = two hydrogens (H) and one oxygen (O). The 2 belongs to the H, not the O.

3. D - One carbon (C) plus two oxygens (O_2) = 3 atoms. No subscript means 'one'.

4. C - Air is about 21% oxygen and 78% nitrogen. Room air is why NICU oxygen settings start at '21%' — that's plain air, before any extra is added.

5. B - Nitrogen is about 78% of air. Your lungs take it in and let it out unchanged; only the oxygen gets used.

6. D - That two-way swap across the alveoli walls is gas exchange: fresh oxygen in, waste carbon dioxide out, with every breath.

7. C - Carbon dioxide dissolved in blood makes acid. A little is normal; too much (when breathing fails) makes the blood too acidic. That's why NICU teams measure it in a blood gas test — coming up in Week 2.

8. A - Priestley's 'pure air' kept the mouse alive longer and made a candle burn brighter — because it was pure oxygen, the one-fifth of air that matters. (He also breathed some himself and said it felt 'light and easy.')

Biology 101 — Week 1 (pass mark 6/8)

1. A - Cells are the smallest units that are alive. (The smallest building block of ALL matter is the atom — that's chemistry. Cells are made of billions of atoms.)

2. C - A monk's little bedroom was called a cell, and the cork's rows of tiny boxes looked like a monastery hallway to Hooke. The name stuck for all of biology.

3. B - Mitochondria are where the oxygen you breathe finally gets used — combining with sugar to release energy. The nucleus holds instructions; the membrane is the skin; the cytoplasm is the jelly.

4. B - Doubling: 1 -> 2 -> 4 -> 8. Division doesn't add one cell each time — it doubles the whole crowd. That's how one cell becomes a baby with trillions.

5. D - Red blood cells throw out their nucleus to become pure cargo ships for hemoglobin. Without a nucleus they can't repair themselves — so they wear out after about four months and get recycled (hello, bilirubin).

6. A - Cells team up into tissues (like muscle tissue); tissues team up into organs (like the heart). Organs team up into systems, and systems make an organism — you.

7. D - Skin is a wall built of cell layers, and a preemie's wall is only a few bricks thick. That's the cell-level reason the isolette has to be so warm and humid.

8. A - Roughly 30 trillion — 30,000,000,000,000. And every one of them descends from a single cell by division after division after division.

Physics 101 — Week 1 (pass mark 6/8)

1. D - Pressure = push ÷ area. Spread a push over a big area and the pressure is low; concentrate it on a tiny area and the pressure is huge.

2. C - Same push, far smaller area -> far higher pressure. That's why the sharp end goes in and the flat end doesn't hurt your thumb.

3. D - A blood pressure of 60 mmHg means: enough push to lift a column of mercury 60 millimeters high. Old blood pressure meters literally had a mercury column inside.

4. A - The weight of the whole atmosphere pushing on the dish balanced 760 mm of mercury. That's why '760 mmHg' is normal air pressure — and why blood pressure borrowed the unit.

5. B - Centimeters of water: the gentle pressures of breathing support are measured by how high they could lift water. In bubble CPAP, the tube's depth under water literally sets the pressure.

6. C - Mercury is about 13.6 times heavier than water, so 1 mmHg ~ 1.36 cmH_2O, and 40 mmHg ~ 54 cmH_2O. Bigger unit, bigger pressure — even though millimeters are smaller than centimeters.

7. D - 760 mmHg — Torricelli's column. Your body is pressed on by that much air all the time; you don't feel it because the pressure inside you pushes back equally.

8. B - Pressure can heal and pressure can hurt. Overstretched alveoli scar (a lung disease called BPD) or burst (an air leak). 'Gentle ventilation' is one of the most important ideas in modern NICU care.

Calculus 101 — Week 1 (pass mark 6/8)

1. D - Two big questions: how fast is something changing (rate), and how much has accumulated (total)? A baby's growth chart asks both.

2. D - Change = 970 - 900 = 70 g. Time = 8 - 1 = 7 days. Rate = 70 ÷ 7 = 10 grams per day.

3. C - Steepness (slope) IS the rate. Steep = big change each day; gentle = small change each day.

4. C - Flat = zero change per day. For a preemie who is supposed to be growing, a flat week is a red flag the team acts on.

5. A - Downhill = negative rate. Losing up to about 10% of birth weight in the first days is normal (mostly water), and the line turns uphill again within a week or two.

6. C - Newton in England and Leibniz in Germany each invented it in the late 1600s. Their supporters fought for decades about who was first. (Answer: both.)

7. B - 'Per kilogram per day' lets doctors compare babies of different sizes. A 2 kg baby gaining 40 g/day and a 1 kg baby gaining 20 g/day are both at 20 g/kg/day — both growing well.

8. B - One point can lie (extra fluid, a wet diaper, a different scale). A line through many points tells the truth. That's calculus thinking: look at the change, not the snapshot.

Statistics 101 — Week 1 (pass mark 6/8)

1. D - One number is a fact; a thousand numbers are a mess — until statistics. It's the toolkit for asking 'what's typical?' and 'is this one unusual?'

2. B - Add them up: 725. Divide by how many: 725 ÷ 5 = 145. (725 is the total, not the mean.)

3. D - Range = highest - lowest = 160 - 130 = 30. It tells you how spread out the numbers are.

4. A - Five signs, each scored 0, 1, or 2, at one minute and five minutes after birth. Before Apgar, there was no standard way to say how a newborn was doing — just opinions.

5. A - 7–10 is reassuring; 4–6 means the baby may need some help; 0–3 means the baby needs help right now. Most healthy newborns score 8 or 9 (almost nobody gets a 10 — hands and feet are usually still a little blue).

6. D - 'Looks fine' means something different to every doctor. A score means the same thing everywhere — which is what lets you study thousands of babies and learn what works. Turning observations into numbers is where statistics begins.

7. B - An average hides the spread. 7, 7, 7, 7 and 10, 10, 4, 4 have the same mean. That's why statisticians always ask for the range (or more) alongside the average.

8. C - Every 'normal range' in medicine is a statistics result: measure lots of healthy people, find where the middle bunch lands. The machine's alarms are set from that — not the other way around.

Genetics 101 — Week 1 (pass mark 6/8)

1. B - The nucleus is the library. Nearly every cell in the body has a complete copy of the whole book inside its nucleus. (Red blood cells are the famous exception — they threw theirs out.)

2. A - Just four chemical letters — A, T, C, G — repeated about 3 billion times. Everything about how your body is built is spelled with them.

3. A - A gene is a recipe; a protein is the dish. Humans have about 20,000 genes. The 46 volumes are chromosomes; the single letters are A, T, C, G.

4. C - 23 pairs = 46. Each pair has one volume from each parent — which is why you share features with both.

5. D - One egg, one book, split in two — identical DNA. Fraternal twins come from two eggs: they're regular siblings who happened to share a womb.

6. B - Two babies sharing one womb tend to run out of room and arrive early and small — so twins (and triplets) are a big part of any NICU's population.

7. C - Every protein — hemoglobin, surfactant, the enzymes in your mitochondria — is built from a gene's recipe. A typo in the hemoglobin gene causes sickle cell disease (Week 4).

8. D - 'Photo 51,' taken in 1952, showed the X pattern of a helix. Watson and Crick used it to build their famous model in 1953 — and were slow to give her credit.

Psychology 101 — Week 1 (pass mark 6/8)

1. D - Psychology studies minds by studying behavior. (Cells and chemicals: biology. Groups: sociology — this afternoon.)

2. D - Babies vote with their eyes, their mouths, and their hearts. Looking longer, sucking harder, or a heart-rate change all say 'I notice this' — no words needed.

3. A - The babies changed their sucking to 'choose' the story they'd heard in the womb. They remembered it — proof that listening (and learning) starts before birth.

4. B - Hearing switches on around 24–25 weeks. That means every preemie in the NICU can hear — the alarms, the voices, and especially their parents.

5. C - Babies are built to see the face that's feeding them. Beyond that, the world is a blur — and that's fine, because the important stuff is close.

6. C - Als taught NICUs to read a preemie's body language — a splayed hand, a color change, a frown — and to dim lights, cut noise, and time care around the baby's sleep. NICUs look the way they do today partly because of a psychologist.

7. A - Measure, don't assume — again. Once the stress response was measured, pain control for babies changed fast: sugar drops, gentle holding during pokes, and pain scores in the chart.

8. B - A preemie's brain is doing its third-trimester wiring in the NICU. Calm, sleep, and a familiar heartbeat are the conditions that wiring wants — a parent's chest provides all three.

Sociology 101 — Week 1 (pass mark 6/8)

1. C - Psychology zooms in on one mind; sociology zooms out to the group. A NICU is a small society with its own roles, rules, and rituals.

2. A - Nurses are there hour after hour, shift after shift. They notice most problems first — which is why great NICUs make sure nurses are heard.

3. B - NICU pharmacists calculate tiny doses for tiny bodies, where a decimal point in the wrong place could be deadly. The TPN bags from NICU Tutor Lesson 6 come from them.

4. B - A preemie's immune system is weak, and infection kills. Every wiped surface and every washed hand is medicine. The person with the mop is on the team.

5. B - Level I: healthy newborns. Level II: mildly early or mildly sick. Level III: full intensive care for the smallest and sickest. Level IV: everything in III plus complex surgery — the regional center.

6. A - Very small babies survive more often when they're born at a Level III or IV hospital. So systems try to get mothers there before delivery — or move the baby by transport team right after.

7. A - Rituals are sociology's tools for making groups reliable. Same order, every baby, every day: the nurse reports, the resident presents, the pharmacist checks, the family asks. Nothing falls through.

8. D - Hess designed the unit and its incubators; Lundeen ran it and trained generations of nurses. Their partnership is the ancestor of every NICU team today.

Week 1 Final Exam (pass mark 17/24)

1. D - A big idea in your own words. The others are a copied sentence, a list of words with no ideas, and a note about the lesson instead of the content.

2. C - Admitted as a joke, rejected by dozens of schools, avoided by the town — she graduated first in her class in 1849.

3. D - 6 carbons + 12 hydrogens + 6 oxygens = 24 atoms.

4. C - About 78% nitrogen, 21% oxygen. The body mostly ignores the nitrogen — in and right back out.

5. C - The power plants. Their exhaust — carbon dioxide and water — is what you breathe out.

6. A - 1 -> 2 -> 4 -> 8 -> 16 -> 32. Doubling, not adding.

7. D - Pressure = push ÷ area. Tiny area, huge pressure — that's why sharp things go in.

8. B - Heavier liquid, more push per millimeter. That's why gentle CPAP is measured in centimeters of water and blood pressure in millimeters of mercury.

9. B - Change 140 g ÷ 7 days = 20 g/day.

10. D - Flat = zero change per day. For a preemie who should be growing, that's an alarm.

11. A - 540 ÷ 4 = 135. (540 is the total; 30 is the range.)

12. C - 4-6 means 'may need some help.' The score is about the next few minutes, not the next few years — as Dr. Apgar insisted.

13. B - 23 from each, making 46 in 23 pairs.

14. B - Identical twins share the same DNA, so they're always the same sex. A boy-girl pair must be fraternal.

15. D - If you can't ask, measure what they do. Sucking patterns were the babies' vote.

16. A - Developmental care: the third trimester of brain wiring is happening in the NICU, so the NICU tries to act more like a womb.

17. C - Which is why a healthy team culture makes sure nurses are heard.

18. D - Very small babies survive more often at Level III and IV hospitals, so systems move mothers or babies there.

19. B - Gene -> protein (genetics); packed in red cells with no nucleus (biology); grabs O_2 molecules and changes color (chemistry — and the pulse oximeter).

20. C - Calculus follows one thing changing over time; statistics looks across many things at once. Doctors need both.

21. D - The Goldilocks lesson from NICU Tutor Lesson 2 — even oxygen has a right dose.

22. D - The soap-like liquid preemies don't make enough of — NICU Tutor Lesson 5, and Chemistry's soap lesson in Week 3.

23. D - NICU Tutor Lesson 7 — now with Biology's explanation of why red cells wear out (no nucleus, no repairs).

24. A - Sound bounces off bone; the soft spot is nature's door for brain ultrasounds — NICU Tutor Lesson 8.